

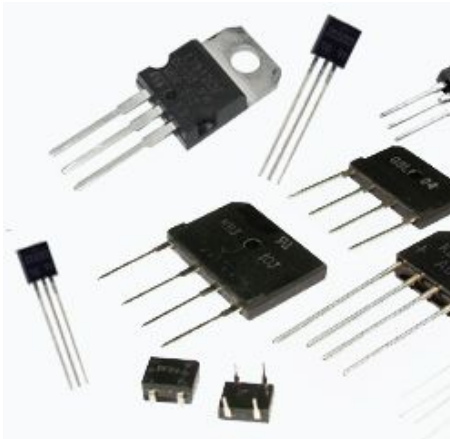


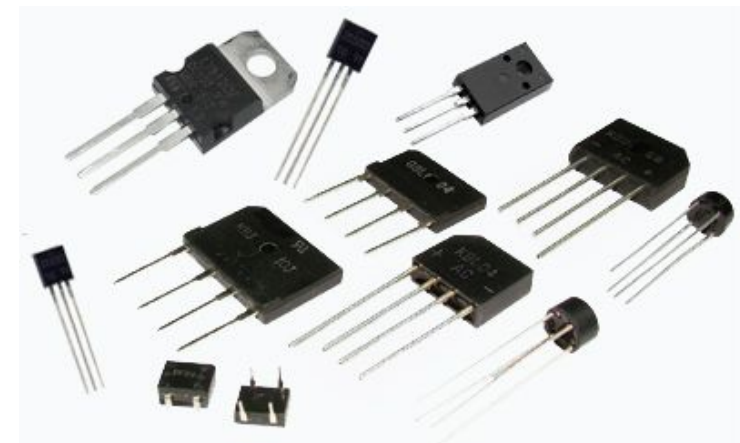
Introduction

Electrical & Electronic Circuits and Elements

Embedded Systems & Embedded Programming

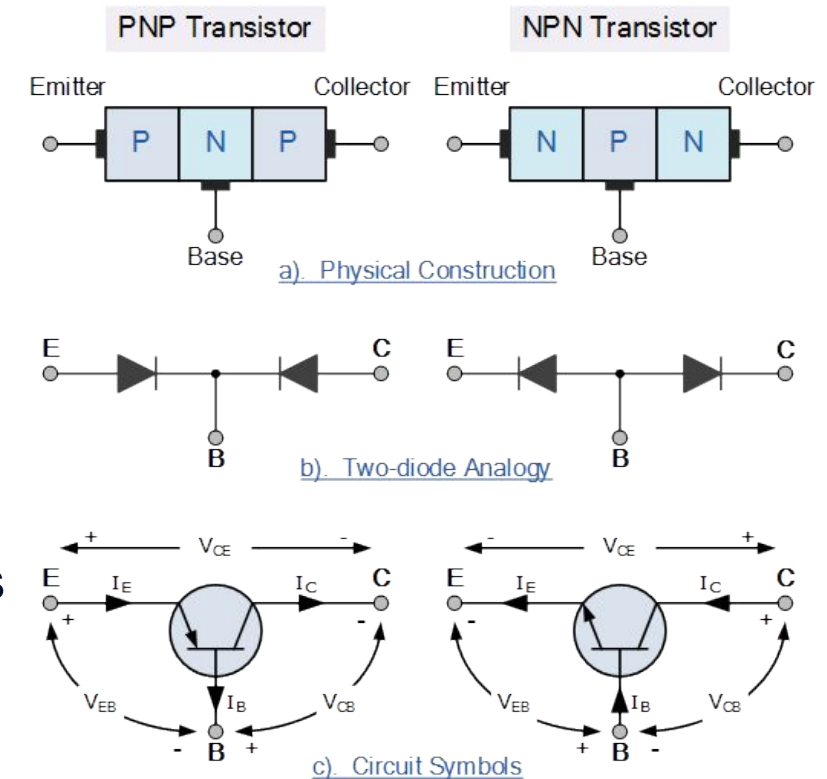
Fundamentals of Electronic Circuits (Transistors)

- ❖ A transistor is an active device that **controls** or **switches** the current flow
 - ❖ Transistor is a combination of two words; **transfer** and **resistor**
 - ❖ They are mainly classified into two types
 - Bipolar Junction Transistor (**BJT**)
 - Field Effect Transistor (**FET**)
 - ❖ They are mainly used
 - As **switches** and signal **amplifiers**
 - In all digital and electronic devices including **ICs**
 - Analog circuits like sensors, RF and etc.
 - ❖ Some IC technologies: **TTL**, **CMOS** and etc.
- 
- A photograph showing a variety of electronic components, including several integrated circuits (ICs) and transistors, scattered on a white background. The components are of different shapes and sizes, with some having multiple pins and others being smaller, discrete components.



Fundamentals of Electronic Circuits (BJT)

- ❖ Bipolar Junction Transistor (**BJT**)
- ❖ Is made by joining together two diodes back-to-back
 - Therefore we have **NPN** and **PNP** types of BJT
- ❖ Both the **electrons** and **holes** are charge carriers
- ❖ Can operate within three different modes:
 - Active (**amplifier**), Saturation (**ON**) and Cutoff (**OFF**)
- ❖ The principle of operation of the **PNP** and **NPN** types is the same and the only difference is biasing and the polarity of the power supply for each type.
- ❖ Was invented by **John Bardeen**, **William Shockley** and **Walter Brattain** in **1948**



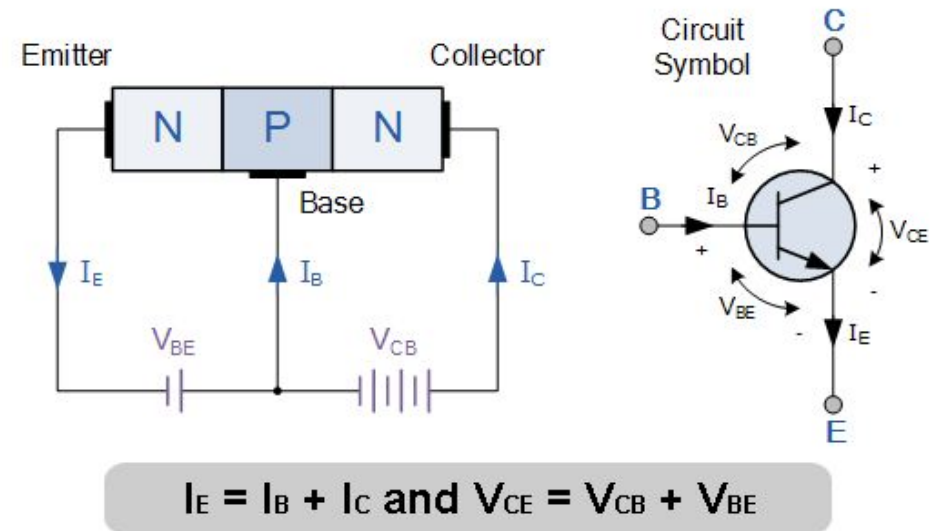
Fundamentals of Electronic Circuits (BJT - NPN)

❖ Cutoff mode

- Both the junctions of the BJT are reverse biased
- No current flows through the device ($I_c = 0$)
- It is in **OFF** state and acts like an **open switch**
- $V_C > V_B < V_E$ ($V_{BE} < 0.7V$ for Silicon)

❖ Saturation mode

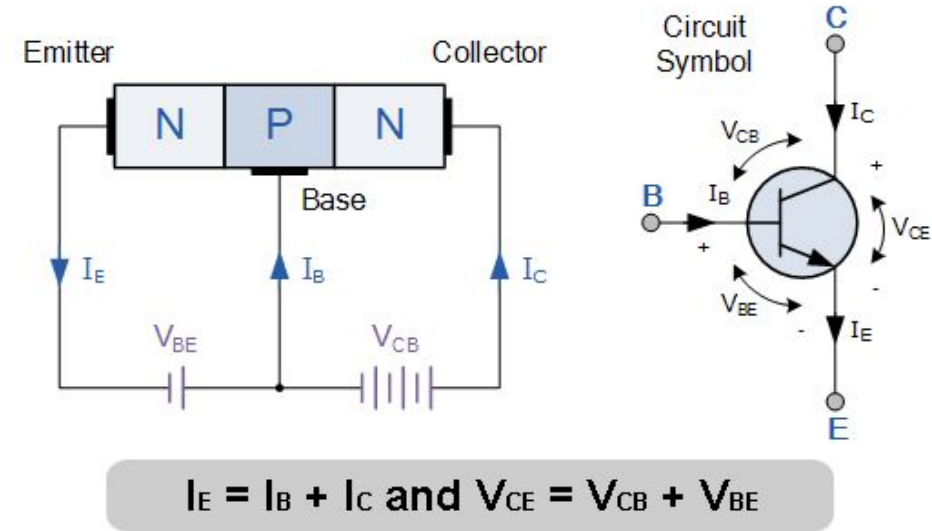
- Both the junctions of the BJT are forward biased
- Huge current flows through the device ($I_c = I_{\text{saturation}}$)
- Increasing I_B doesn't increase I_c ($I_c < \beta * I_B$)
- It is in **ON** state and acts like a **closed switch**
- $V_B > V_C > V_E$ ($V_{BE} > 0.7 V$ for Silicon)
- Collector to emitter saturation voltage $V_{CE(\text{sat})}$ is around 0.05 - 0.2V



Fundamentals of Electronic Circuits (BJT - NPN)

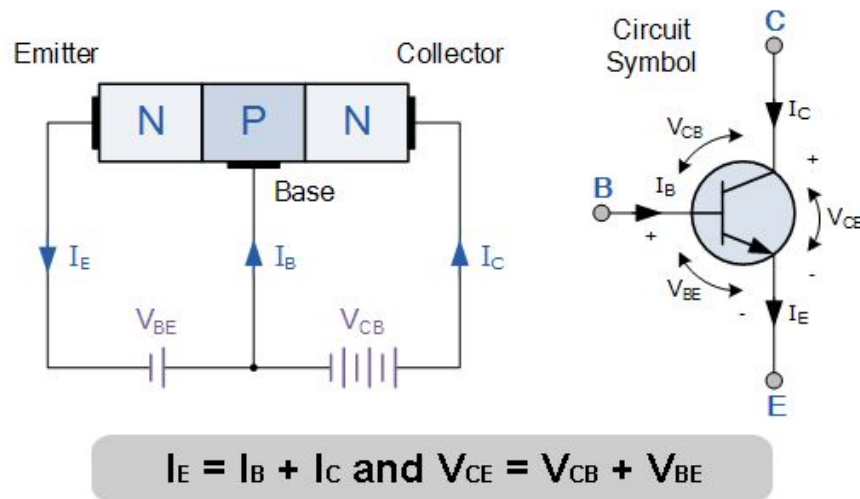
❖ Active mode

- Emitter to base junction is forward biased
- Collector to base junction is reverse biased
- It operates as an **amplifier** and $I_C = \beta * I_B$
- $V_C > V_B > V_E$ ($V_{BE} > 0.7$ V for Silicon)
- β (h_{fe}) is the **DC Current Gain** of the BJT
- β is generally in the range between 50 - 200
- The collector to emitter current gain is called Alpha, $I_C = \alpha * I_E$
- α and β are specified in the datasheet of the device
- $\alpha = \beta / (\beta + 1)$



Fundamentals of Electronic Circuits (BJT - NPN)

❖ i-v characteristic of BJT



[NPN Transistor](#)



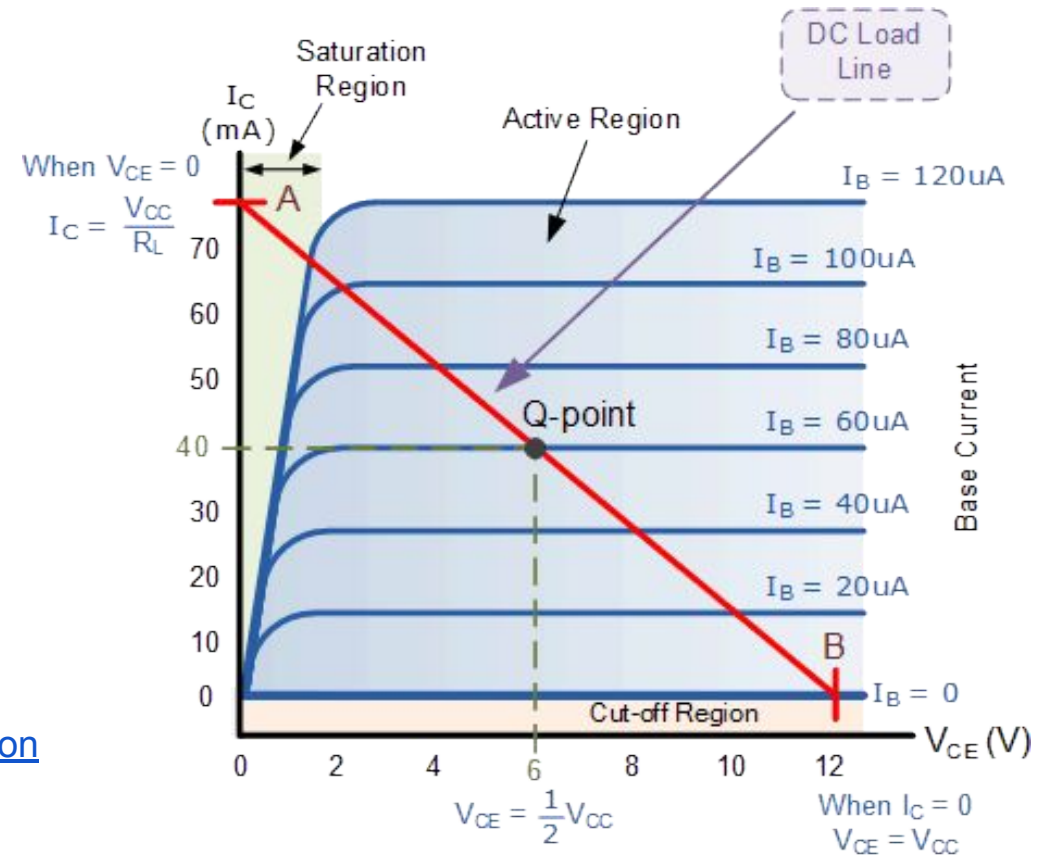
[How does a Transistor Work? A Simple Explanation](#)



[Transistors, How do they work?](#)



[Transistors Explained - How transistors work](#)



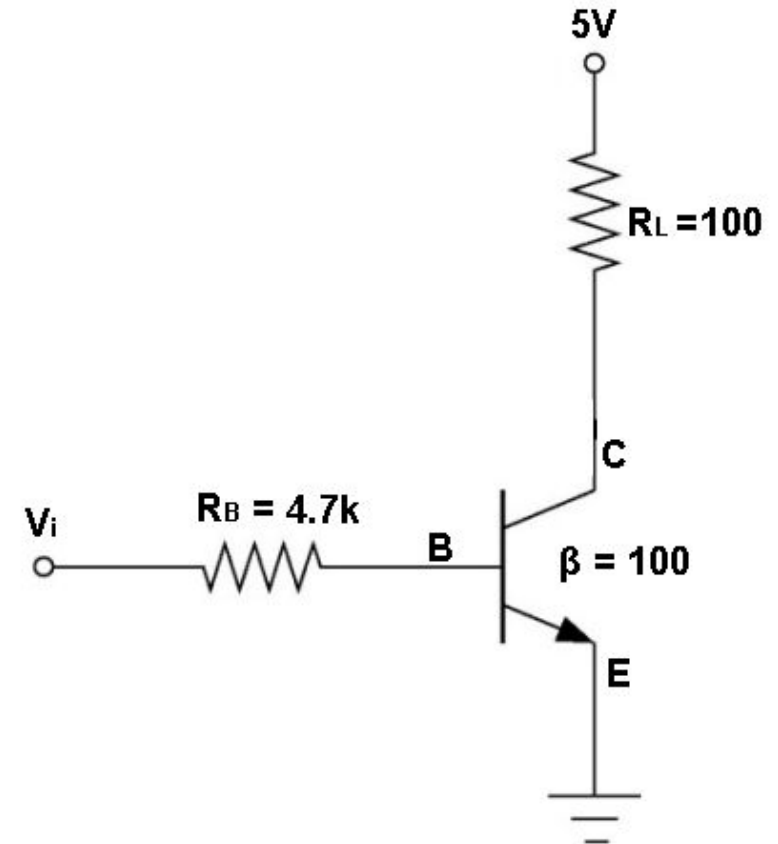
BJT Operating Regions

Fundamentals of Electronic Circuits (BJT - NPN)

- ❖ Find the operation **mode**, **I_B**, **I_C**, **I_E**, **V_{BC}** and **V_{CE}** if
 - $V_i = 0.5 \text{ V}$ (Cutoff, 0 A, 0 A, 0 A, -4.5 V, 5 V)
 - $V_i = 2.0 \text{ V}$ (Active, 0.28 mA, 28 mA, 28.28 mA, -1.5 V, 2.2 V)
 - $V_i = 4.0 \text{ V}$ (Saturation, 0.7 mA, 50 mA, 50.7 mA, 0.7 V, 0 V)
- ❖ Find the minimum V_i and I_B to get the BJT saturated
 - $V_i = 3.05 \text{ V}$
 - $I_B = 0.5 \text{ mA}$

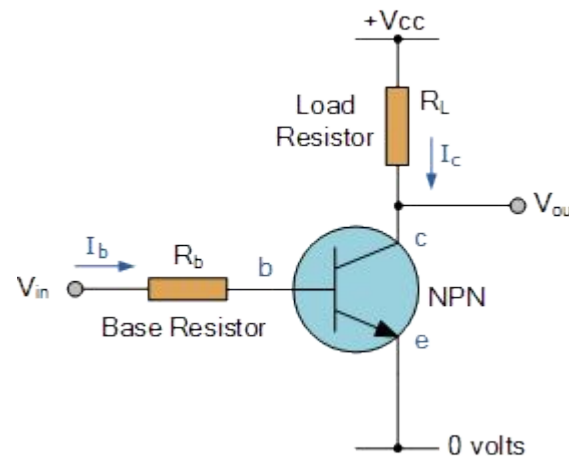
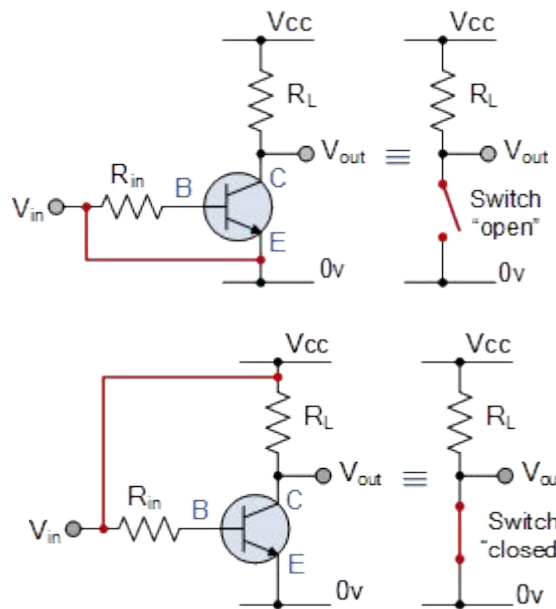
KCL: $I_E = I_B + I_C$

KVL: $V_{CE} = V_{CB} + V_{BE}$



Fundamentals of Electronic Circuits (BJT - NPN)

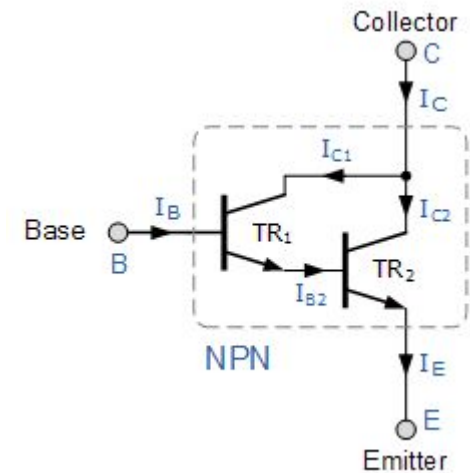
- ❖ Darlington Transistor (BJT as amplifier)
 - When large currents or voltages need to be controlled
- ❖ BJT as a switch
 - Can be used to switch and control lamps, relays or even motors



[Transistors](#)



[Transistors as a Switch](#)



$$I_C = (\beta_1 + \beta_2 + \beta_1 \cdot \beta_2) \cdot I_B$$

$$\beta = (\beta_1 + \beta_2 + \beta_1 \cdot \beta_2)$$

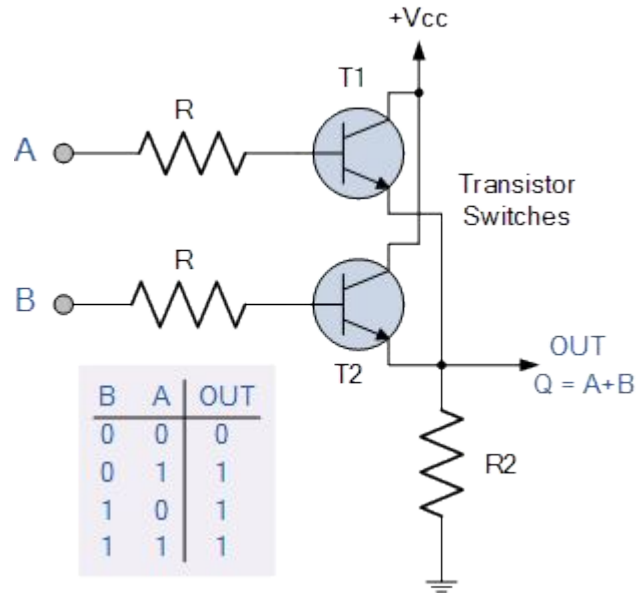
$V_{BE} > 1.4 \text{ V}$ (for silicon)



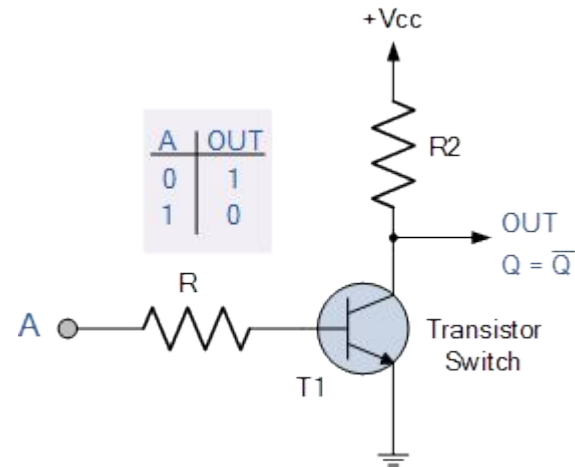
[Transistor as a Switch](#)

Fundamentals of Electronic Circuits (BJT - NPN)

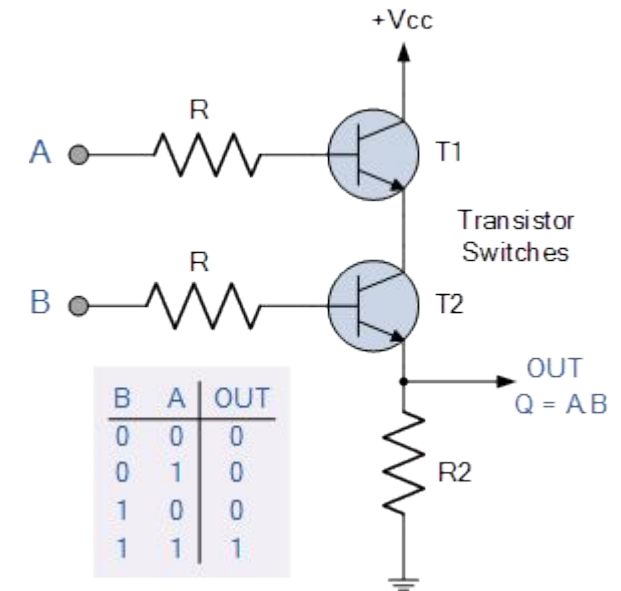
❖ BJTs and Logic Gates



Logic OR Gate



Logic NOT Gate



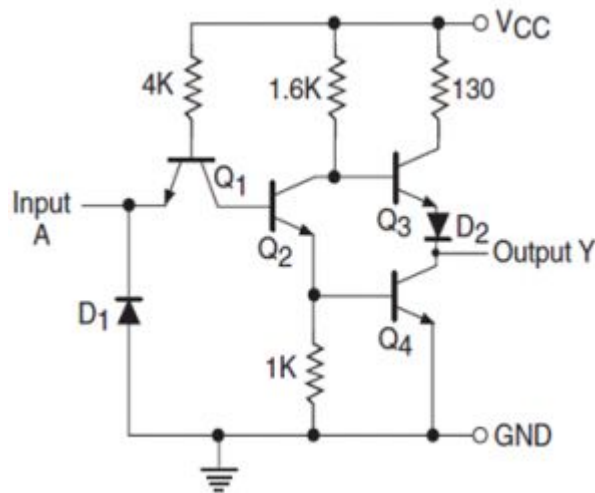
Logic AND Gate



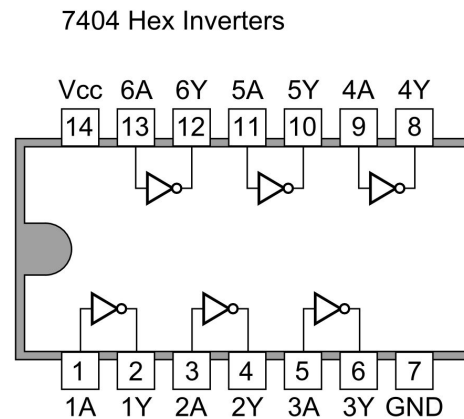
Fundamentals of Electronic Circuits (BJT - NPN)

❖ TTL (Transistor–transistor logic) family

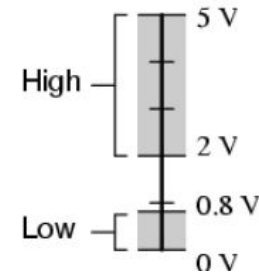
- Is a logic family built from bipolar junction transistors
- Introduced in integrated circuit form in 1963 by Sylvania
- The **7400** series by Texas Instruments became particularly popular



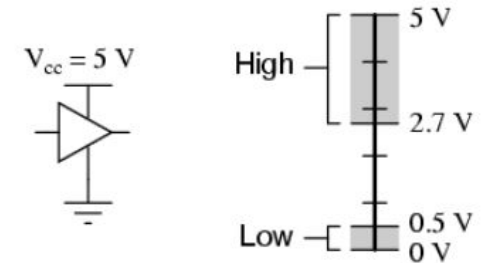
TTL Inverter (NOT) IC



Acceptable TTL gate input signal levels



Acceptable TTL gate output signal levels



Fundamentals of Electronic Circuits (Field Effect Transistors)

❖ Field Effect Transistor (**FET**)

- Has three terminals (**Source**, **Gate** and **Drain**)
- Can be used as switch, amplifier, resistor, capacitor and etc.
- Is high efficiency and robust and can be used in most applications to replace **BJTs**
- Is a unipolar semiconductor device (charge carriers are electrons **or** holes)
- Unlike **Bipolar Junction Transistor**, it is controlled by an electric field (**Gate Voltage**)
- Two basic classifications: the **N-channel FET** and the **P-channel FET**
- Has a very high input impedance (more than $10^9\Omega$)
 - Very sensitive to input voltage signal and can be easily damaged by static electricity
 - Lower power consumption and smaller than an equivalent BJT transistor
 - Very low gate leakage current (less than 10^{-9}A)
- Types of FET: JFET, **MOSFET**, MESFET, FinFET and etc.



[Field Effect Transistors](#)

Fundamentals of Electronic Circuits (Field Effect Transistors)

❖ Field Effect Transistor (**FET**)

- It has a high power gain and It can deliver more power than BJT
- It is less noisy and has a higher frequency response in compare with BJT
- Has a very low "ON" resistance and have a high "OFF" resistance
- It can dissipate large amounts of power while switching
- It provides more thermal stability
- Applications
 - FETs are widely used in analog and digital circuits
 - Power converters and suppliers, RF power amplifiers and etc.
 - Integrated circuits like processors such as the **CMOS** range of digital logic chips
 - Sensors; like pressure and image sensors
 - Volatile and non-volatile memories



 [Field Effect Transistor Types & Technology](#)

Fundamentals of Electronic Circuits (MOSFET)

❖ Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

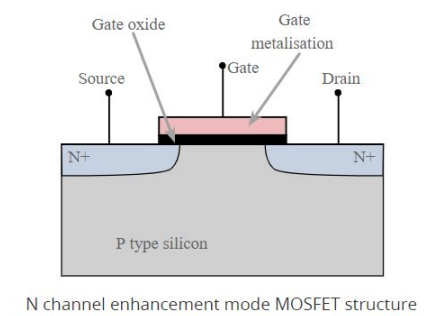
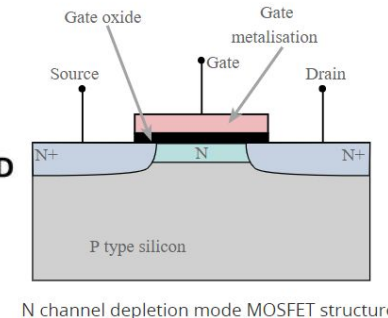
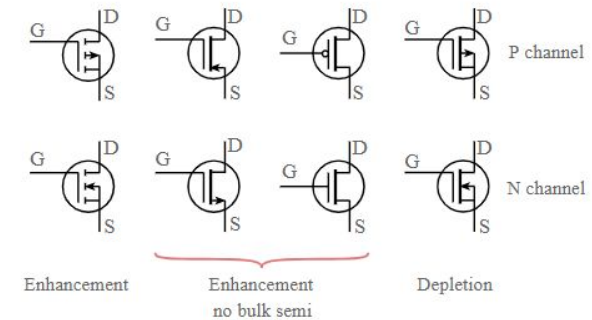
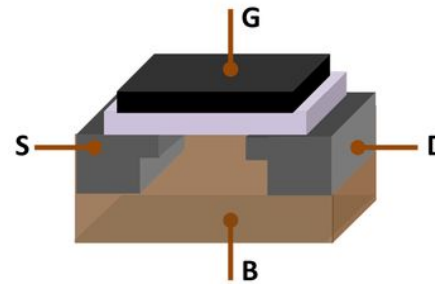
- The gate is physically insulated from the channel by an oxide layer (SiO_2)
- Voltage applied to the gate controls the conductivity of the channel
 - As a result of the electric field induced capacitively across the insulating dielectric layer



- Both N-channel (NMOS) and P-channel (PMOS) variants are available
- Both enhancement and depletion types are available
- Typically offers more than $10^{14} \Omega$ of input impedance
- Is the most manufactured device in history
 - 1.3×10^{22} MOSFETs between 1960 and 2018

➤ Three terminals

- Gate, Drain, Source
- Bulk/Body is connected to source or has no effect (in no bulk E-MOSFET)



Fundamentals of Electronic Circuits (MOSFET)

❖ Depletion MOSFET

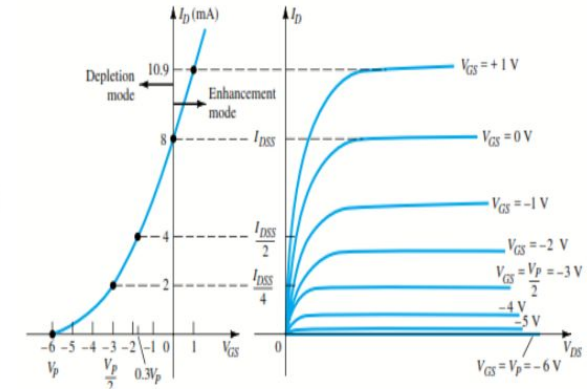
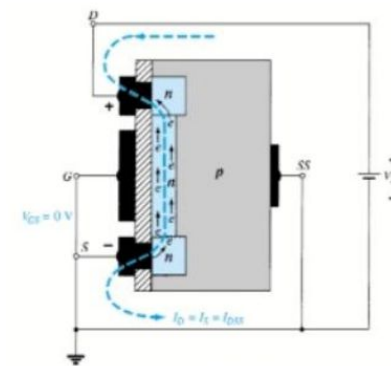
- It has a diffused channel between the drain to source terminal
- It requires the Gate-Source voltage, (V_{GS}) to switch the device “OFF”
- It is equivalent to a “Normally Closed” switch
- it is possible to bias its gate in either polarity, positive or negative



[Depletion Type MOSFET: What is it?](#)

■ For example in a N-channel MOSFET

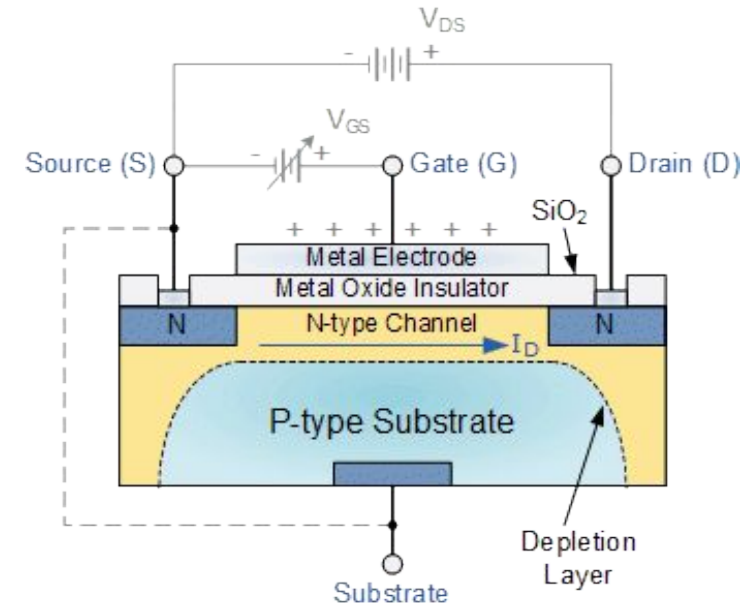
- More $+V_{GS}$ means more current
- While more $-V_{GS}$ means less current
- Operate within three different regions
 - Cut-off, Linear (Ohmic) and Saturation
- Is less common than the enhancement mode types
- It is specified in term of I_{DSS} and V_P
$$i_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$



Fundamentals of Electronic Circuits (MOSFET)

❖ Enhancement MOSFET (eMOSFET)

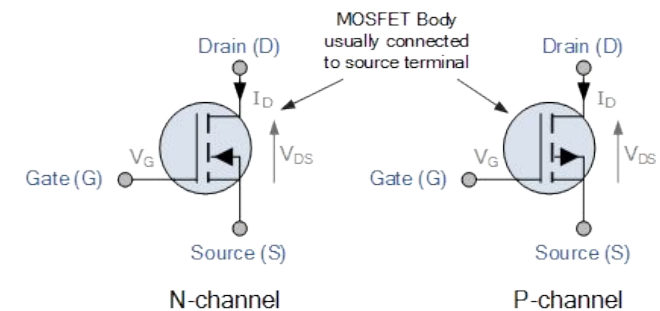
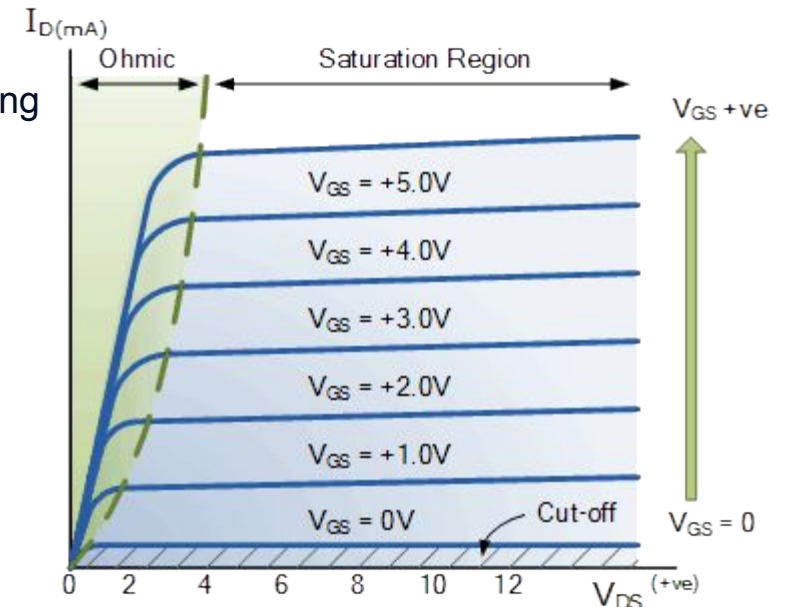
- The conducting channel is lightly doped or even undoped
- The device is normally “OFF” when V_{GS} is 0V
- It requires a Gate-Source voltage to switch the device “ON”
- It is equivalent to a “Normally Open” switch
- For an n-channel eMOSFET
 - + V_{GS} turns it “ON”, while a zero or - V_{GS} turns it “OFF”
 - Increasing the + V_{GS} will decrease the channel resistance
- A drain current will only flow when V_{GS} greater than the threshold voltage (V_{TH})
- It is an excellent electronics switch due to its
 - Infinitely high input impedance
 - Low “ON” and extremely high “OFF” output impedances
- eMOSFETs are used in ICs to produce CMOS type Logic Gates and power switching circuits



[MOSFET Transistor Basics & Working Principle](#)

Fundamentals of Electronic Circuits (MOSFET)

- ❖ An eMOSFET is specified in term of (V_{th}) and K_n
 - V_{th} (threshold) is the minimum required V_{GS} to make the device conducting
 - K_n is the conductance coefficient and it can be calculated from
 - $I_{D(on)}$ and $V_{GS(on)}$ form its datasheet
- ❖ eMOSFET operation modes
 - Cut-off Region ($V_{GS} \leq V_{th}$):
 - It is off thus, $I_D = 0$
 - It acts like an open switch regardless of the value of V_{DS}
 - Ohmic Region ($V_{GS} > V_{th}$ and $V_{DS} \leq V_{GS} - V_{th}$):
 - A voltage-controlled resistance
 - It acts like a closed switch (V_{DS} very small)
 - Saturation Region ($V_{GS} > V_{th}$ and $V_{DS} > V_{GS} - V_{th}$):
 - High V_{DS} and a high I_D simultaneously



Fundamentals of Electronic Circuits (MOSFET)

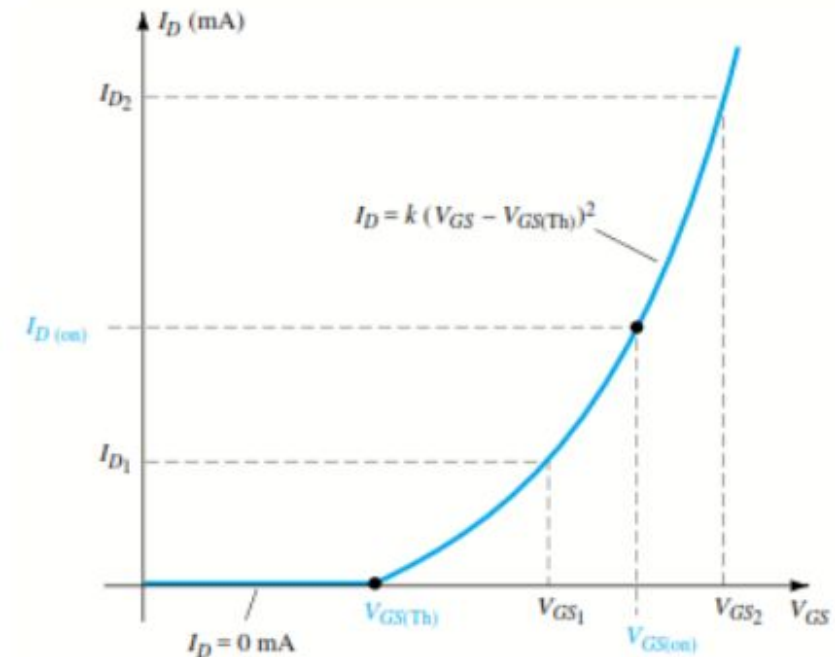
- ❖ The conductance coefficient is $k_n = \mu_n C_{ox} \frac{W}{L}$
 - μ_n is the charge-carrier effective mobility, C_{ox} is the gate oxide capacitance per unit area
 - W is the gate width and L is the gate length
- ❖ The drain current in ohmic mode is
 - $I_D = K_n(V_{GS} - V_{th} - 0.5V_{DS})V_{DS}$
- ❖ The drain current in saturation mode is
 - $I_D = 0.5K_n(V_{GS} - V_{th})^2$
- ❖ The gain of an eMOSFET in saturation mode is
 - $g_m = 2 \times K_n \times (V_{GS} - V_{th})$



[MOSFETs and How to Use Them | AddOhms](#)



[Power Electronics - MOSFET Power Losses](#)



Fundamentals of Electronic Circuits (MOSFET)

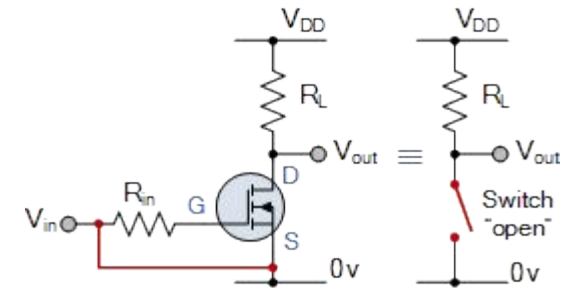
❖ MOSFET as a switch

➤ OFF State

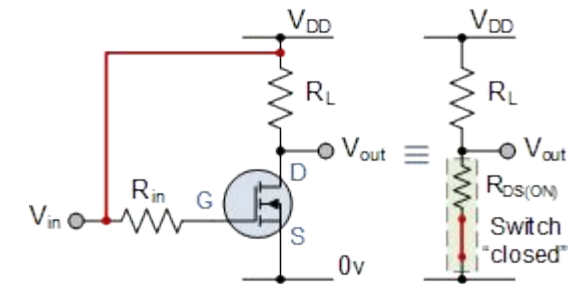
- $V_{GS} = 0V$ and $V_{GS} < V_{TH}$
- MOSFET is “OFF” and $I_D = 0A$
- MOSFET operates as an “open switch”
- $V_{OUT} = V_{DS} = V_{DD} == \text{HIGH}$

➤ ON State

- $V_{GS} = V_{DD}$ and $V_{GS} > V_{TH}$
- MOSFET is “ON” and $I_D = V_{DD} / R_L$
- $V_{DS} = 0V$ and the channel resistance $R_{DS(on)} < 0.1\Omega$
- MOSFET operates as a low resistance “closed switch”
- $V_{OUT} = V_{DS} \approx 0.2V$ due to $R_{DS(on)}$



The MOSFET is in the cut-off mode



The MOSFET is in saturation mode

Fundamentals of Electronic Circuits (MOSFET)

❖ Example: Calculate V_{GS} , V_{DS} and I_D

❖ Known values

➤ $V_{GG} = 1.5V$, $V_{DD} = 7V$, $V_{th} = 1V$, $K_n = 1.2mA/V^2$ and $R_D = 7\text{ k}\Omega$

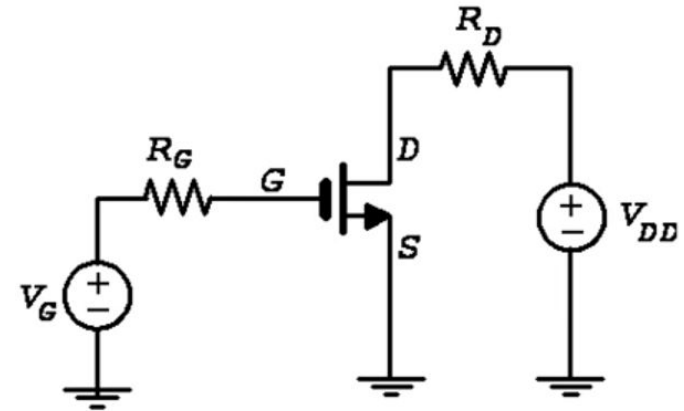
❖ Answer

➤ $I_G = 0$, $V_{GS} = V_{GG} = V_G - V_S = 1.5V$

➤ $V_{GS} > V_{th}$, we assume that the MOSFET is in saturation mode

➤ $I_D = 150\text{ }\mu A$ and $V_{DS} = 5.95V$

➤ $V_{DS} > V_{GS} - V_{th} \Rightarrow 5.95 > 1.5 - 1$  so the transistor is in the saturation mode



 [The MOSFET](#)

 [MOSFET as a Switch](#)

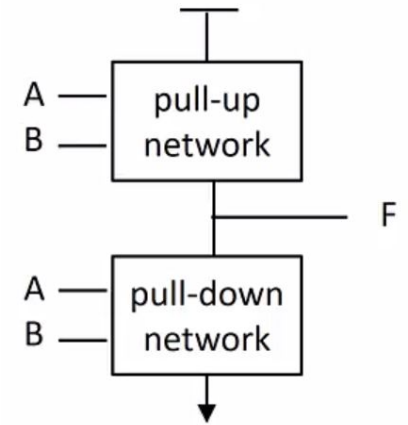
Fundamentals of Electronic Circuits (CMOS)

❖ CMOS (Complementary MOS)

- Is a type of MOSFET fabrication process
- Uses complementary and symmetrical pairs of NMOS and PMOS for logic function

❖ Features

- Dissipates low power
 - At 1 MHz and 50 pF load, the power dissipation is typically 10 nW per gate
 - Power consumption increases with higher clock speeds
- Short propagation delays (around 25 ns)
- Good noise immunity (50% or 45% of the full logic)
 - $NM_H = NM_L = 3(V_{DD} + 2V_{th}/3)/8$
- Levels of the logic signal will be equal to V_{DD} since the input impedance is so high.
- Voltage levels range from 0 to V_{DD}
- A low level is between 0 and $1/3 V_{DD}$ while a high level is between $2/3 V_{DD}$ and V_{DD}



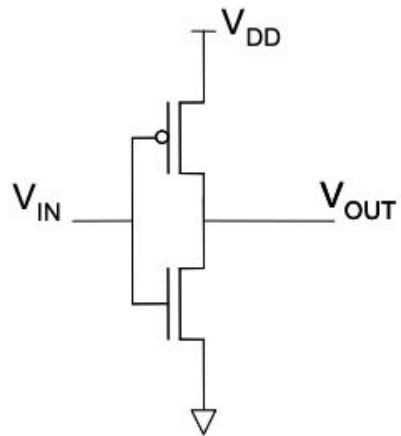
Pull-up: PMOS only
Pull-down: NMOS only

Fundamentals of Electronic Circuits (CMOS)

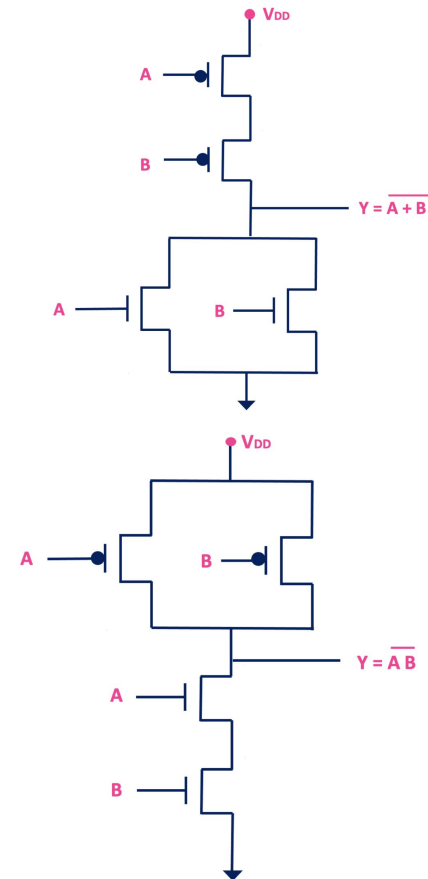
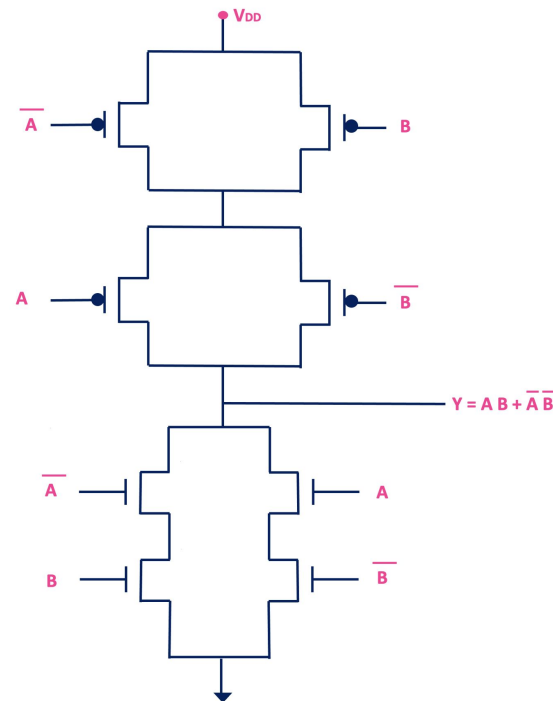
❖ Basic Logic Gates

❖ Useful links

- [CMOS Logic Gates](#)
- [CMOS Logic Gates](#)
- [CMOS Transistors](#)
- [CMOS Introduction](#)



XNOR Gate



Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Fundamentals of Electronic Circuits

❖ Some useful links

- [Bipolar Transistor](#)
- [Bipolar Junction Transistor](#)
- [Introduction to Transistors](#)
- [How to Identify an PNP or NPN Transistor](#)
- [Transistor–Transistor Logic \(TTL\)](#)
- [Emitter-coupled logic](#)
- [The Basics of Emitter-Coupled Logic](#)
- [Emitter-Coupled Logic](#)
- [BJTs as Transistor Switches](#)
- [MOSFET - Applications of MOSFET](#)
- The MOSFET ([part 1](#), [part 2](#) and [part 3](#))
- [MOSFET](#)
- [Depletion MOSFETs](#)
- [Using Power MOSFETS with Arduino](#)